Clients guide tests we perform in the laboratory

Customer-facing catalogue of laboratory tests and typical execution approach.

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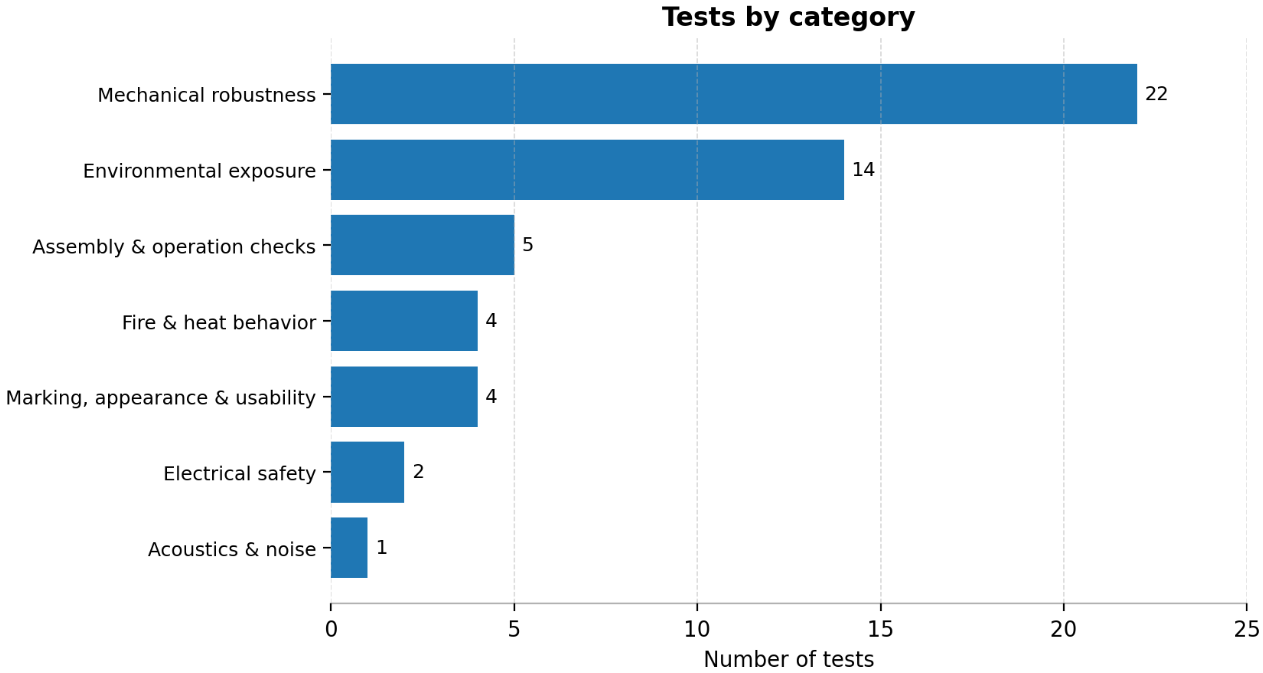
Total tests	52
Categories	7

Overview

The chart below summarizes the distribution of tests across categories.



Category statistics

Category	Number of tests
Mechanical robustness	22
Environmental exposure	14
Assembly & operation checks	5
Fire & heat behavior	4
Marking, appearance & usability	4
Electrical safety	2
Acoustics & noise	1

Standards referenced

Depending on product family, target market and customer requirements, tests may be performed with reference to the following standards.

UL — UL (Underwriters Laboratories) standards and related conformity assessment are widely used for product safety and compliance in the United States and are commonly recognized across North America.

IEC 62208 — IEC 62208 specifies general requirements and test methods for empty enclosures intended for low-voltage switchgear and controlgear assemblies.

IEC 61439-5 — IEC 61439-5 defines specific requirements for low-voltage switchgear and controlgear assemblies for power distribution in public networks (public network distribution assemblies).

IEC 61386-1 — IEC 61386-1 specifies requirements and tests for conduit systems (conduits and fittings) used for cable management and protection in electrical installations.

ISO 13347-1 — ISO 13347-1 provides an overview of standardized laboratory methods used to determine the acoustic performance (sound power) of industrial fans.

Test catalogue

Tests are grouped by category. Each category includes an index table followed by a short description of each test.

Mechanical robustness

Test title	Standard
Impact resistance (IK)	IEC 62208
Static load resistance	IEC 62208
Static load resistance (power distribution)	IEC 61439-5
Shock load resistance	IEC 61439-5
Torsional stress resistance	IEC 61439-5
Door mechanical strength	IEC 61439-5
Crushing resistance (tubes)	IEC 61386-1
Shock resistance (tubes)	IEC 61386-1
Impact resistance	UL
Impact resistance (observation openings)	UL
Crushing resistance	UL
Deflection	UL
Comparative deflection	UL
Mold stress-relief	UL
Misalignment	UL
Gasket tensile & elongation	UL
Gasket compression	UL
Conduit connection (general)	UL
Conduit connection — pullout	UL
Conduit connection — torque	UL
Conduit connection — bending	UL
Conduit connection — breakout	UL

Test descriptions

Impact resistance (IK)

Checks resistance to knocks and impacts

Why it matters: Ensures accidental knocks won't crack the enclosure and expose live parts or break the IP barrier. Example: tool impact during maintenance shouldn't open a gap at the door edge.

How we run it: IK is performed with a calibrated pendulum/impact hammer at defined energy and impact points; after each impact, we inspect cracks, deformation, and verify door closure and gasket compression (gap checks with gauges).

Static load resistance

Validates load handling without damage

Why it matters: Confirms the enclosure won't permanently deform under sustained loads (mounted devices, cable weight, accidental leaning), keeping doors aligned and seals effective.

How we run it: We apply a defined load using calibrated weights or a load frame, measure deflection with a dial gauge/LVDT, then check for permanent set and correct operation of doors/locks.

Static load resistance (power distribution)

Mechanical strength under static load

Why it matters: Ensures heavy distribution assemblies don't distort mounting planes, which can stress busbar supports and compromise clearances/creepage distances.

How we run it: Similar static loading but with representative load locations; we measure deflection at critical points (mounting rails, base) and verify door alignment and internal clearances with calipers and gauges.

Shock load resistance

Checks behavior under sudden load

Why it matters: Simulates sudden forces during transport/handling so the enclosure still protects devices after a drop-like event or abrupt impact.

How we run it: We use a controlled shock/drop test rig (defined mass/height or acceleration pulse). After shocks, we inspect structure, verify closures, and check that no sharp edges or openings appear.

Torsional stress resistance

Checks twisting resistance

Why it matters: Ensures twisting (mounting on uneven walls, handling) won't break gasket lines or misalign doors—protecting IP and safe access.

How we run it: The enclosure is clamped in a torsion fixture; we apply a controlled torque using a torque wrench/actuator, measure twist/deflection, then confirm door operation and gasket contact.

Door mechanical strength

Validates door robustness

Why it matters: Doors are the most-used interface; fatigue or hinge failure can lead to poor sealing, unsafe access, or broken locks.

How we run it: We run repeated opening/closing cycles on a cycling rig, measuring operating forces with a force gauge, checking hinge wear, alignment, latch engagement, and gasket compression over the test.

Crushing resistance (tubes)

Checks resistance to crushing forces

Why it matters: Ensures protective tubes/conduits won't collapse and expose or pinch cables, which can damage insulation and create faults.

How we run it: Tubes are compressed on a universal testing machine or compression rig; load is measured and displacement monitored, then inspected for cracking and loss of protective function.

Shock resistance (tubes)

Checks resistance to shocks

Why it matters: Confirms tubes resist accidental hits during installation—so cable protection remains intact.

How we run it: We apply defined impacts using a calibrated impact tester (drop weight/hammer), then inspect for cracks and check that tube geometry still allows safe cable routing.

Impact resistance

Checks resistance to impacts

Why it matters: Adds compliance confidence for markets referencing UL methods—ensuring consistent enclosure toughness and safe containment after impact.

How we run it: Per UL setup, we apply impacts at specified locations using the defined striker; we verify no unsafe openings, no access to live parts, and continued closure function.

Impact resistance (observation openings)

Checks openings remain safe after impact

Why it matters: Viewing windows must not shatter or detach—avoiding sharp fragments and preventing access to energized parts.

How we run it: We impact the window area using a calibrated striker; afterwards we check retention, cracking pattern, and that the opening still prevents finger/probe access where required.

Crushing resistance

Checks resistance to crushing forces

Why it matters: Confirms the enclosure can withstand compressive forces (stacking, accidental pressure) without creating gaps that defeat IP or damage internal devices.

How we run it: We apply compressive load with a press/load frame, measure deformation, then assess permanent set and closure/sealing performance.

Deflection

Checks deformation under load

Why it matters: Limits bending so doors still close correctly and clearances to live parts are maintained—critical for electrical safety and IP performance.

How we run it: A defined load is applied; deflection is measured using a dial indicator/LVDT at specified points, with pass/fail against allowed limits.

Comparative deflection

Compares deformation performance

Why it matters: Ensures consistent stiffness across variants (sizes/materials), so installation behavior is predictable for customers.

How we run it: Same deflection method, but repeated across variants under identical load points; results are recorded and compared to design targets.

Mold stress-relief

Assesses stability after molding

Why it matters: Detects residual molding stresses that can later warp parts, causing poor gasket compression and IP failures months after installation.

How we run it: Molded parts are conditioned in a controlled oven cycle; we measure dimensions before/after with calipers and check for warpage/cracks and assembly fit.

Misalignment

Checks fit/closure under stress

Why it matters: Confirms the enclosure still seals and operates when mechanical stresses try to shift the geometry—preventing “almost closed” doors that leak water/dust.

How we run it: Under defined loading/torsion, we check alignment using gauges, calipers, and functional closure tests (latch engagement, uniform gasket compression).

Gasket tensile & elongation

Assesses gasket material robustness

Why it matters: Ensures gaskets won’t tear when stretched during assembly or door cycling—keeping the IP barrier intact over years.

How we run it: We cut standardized specimens and test on a tensile machine; force/elongation are recorded to confirm material strength and ductility.

Gasket compression

Checks gasket compression behavior

Why it matters: Confirms the gasket maintains sealing force (doesn’t “take a set”)—critical to prevent micro-leaks after months of compression at the door.

How we run it: We run compression-set fixtures at controlled temperature/time, then measure recovery (thickness with calipers) and, if needed, hardness with a durometer.

Conduit connection (general)

Checks robustness of conduit connection

Why it matters: Ensures cable/conduit entries remain mechanically secure so sealing isn’t lost and cables aren’t stressed—protecting IP and electrical safety.

How we run it: We mount fittings per instructions and verify mechanical robustness through defined pull/torque/bending checks; measurements use force gauges and torque tools with documented limits.

Conduit connection — pullout

Checks resistance to pulling out

Why it matters: Prevents accidental pull from loosening glands and creating leak paths or stressing terminals.

How we run it: We apply an axial tensile force using a calibrated dynamometer/test stand, record peak load, and inspect for movement, cracking, or sealing loss.

Conduit connection — torque

Checks resistance to twisting torque

Why it matters: Ensures tightening or twisting the conduit won't crack the wall or loosen the seal.

How we run it: We apply specified torque with a calibrated torque wrench, then inspect threads, wall integrity, and sealing interface.

Conduit connection — bending

Checks resistance to bending forces

Why it matters: Confirms side loads from cables/conduits won't open gaps or crack the entry zone.

How we run it: A defined bending load is applied using a fixture and measured with a force gauge; we then inspect for cracks and verify entry retention and sealing.

Conduit connection — breakout

Checks breakout behavior

Why it matters: Ensures knockouts/breakouts don't weaken the wall excessively—so once opened, the enclosure still withstands mechanical stresses without cracking.

How we run it: We remove breakouts as intended, then apply defined mechanical stresses around the opening and inspect for crack propagation; dimensional checks use calipers.

Environmental exposure

Test title	Standard
Ingress protection (IP)	IEC 62208
UV resistance	IEC 62208
Corrosion resistance	IEC 62208
Rain test	UL
Drip test	UL
Dust test (hose method)	UL
Atomized water (method A)	UL
Atomized water (method B)	UL
Hose-down test	UL
Indoor corrosion protection	UL
Additional corrosion protection	UL
Outdoor corrosion (salt spray 60h)	UL
Oil exclusion	UL
Oil immersion	UL

Test descriptions

Ingress protection (IP)

Checks dust/water tightness level

Why it matters: Confirms the enclosure really protects electrical devices against dust and water for its declared IP rating—reducing risks like insulation tracking, corrosion, or short circuits after wash-down or rain. Example: preventing water reaching terminals during outdoor cabinet operation.

How we run it: We test in controlled setups per the IP code: dust chamber (standardized dust + airflow, often with vacuum draw), and water tests using calibrated nozzles, pump, flowmeter, and pressure gauge, with timed exposures and defined angles/turntable; then we inspect ingress and functionality.

UV resistance

Ensures outdoor sunlight durability

Why it matters: Ensures plastics/coatings won't embrittle, crack, or lose mechanical strength outdoors—so doors, latches, and seals keep protecting live parts over years. Example: preventing a sun-aged door from warping and breaking the gasket line.

How we run it: Samples (or critical parts) go into a UV weathering chamber with controlled irradiance, temperature, and cycles; we monitor with an irradiance sensor and black-panel thermometer, then assess cracking, color change, and mechanical integrity (visual + dimensional checks with calipers).

Corrosion resistance

Helps prevent rust and degradation

Why it matters: Validates coatings/materials protect hinges, locks, fasteners, and bonding points—avoiding seized doors, compromised PE connections, or loss of sealing over time. Example: rust on a hinge can misalign the door and create water paths.

How we run it: We expose parts/enclosures in salt spray (fog) chambers or controlled humidity/corrosive atmosphere chambers; key parameters are logged (temperature, pH, salinity). After exposure, we rate corrosion, check coating adhesion, and measure any critical dimensions with calipers.

Rain test

Simulates rainfall exposure

Why it matters: Confirms rain won't reach live parts through door joints, roof interfaces, or cable entries—critical for outdoor cabinets protecting breakers, terminals, and control devices.

How we run it: We use a rain simulation rig (calibrated flow, nozzle pattern, angles) with timed exposure; water flow/pressure is verified using flowmeter/pressure gauge. Post-test: internal inspection for droplets/tracks and functional checks.

Drip test

Checks resistance to dripping water

Why it matters: Ensures vertically dripping water (condensation, overhead leaks) won't reach terminals or insulation—preventing nuisance trips, corrosion, or tracking in indoor/sheltered installs.

How we run it: We test under a drip box with controlled droplet rate and height, using a leveled fixture; duration is timed and verified. After exposure, we check for ingress paths and moisture near critical components.

Dust test (hose method)

Simulates dust ingress

Why it matters: Confirms wind-driven or industrial dust won't accumulate inside—preventing blocked ventilation, abrasive wear, and insulation contamination that can lead to overheating or tracking.

How we run it: We expose the enclosure in a dust airflow rig/tunnel with standardized dust; airflow and time are controlled, and if required we apply slight negative pressure. After test: internal inspection and dust deposit assessment.

Atomized water (method A)

Simulates mist/spray exposure

Why it matters: Validates protection against fine mist/spray (condensation, light splash) that can creep through seams—critical for preserving insulation integrity and preventing corrosion of terminals.

How we run it: We use an atomizing spray chamber with calibrated nozzles and set droplet distribution; exposure time and orientation are fixed. We verify flow/pressure and then inspect for ingress and wetting of critical zones.

Atomized water (method B)

Alternative spray exposure method

Why it matters: Adds confidence under different angles/patterns—useful when cabinets face varied spray during cleaning or installations with complex door geometry.

How we run it: Same spray principle as A but with the method-B orientation/pattern; we keep calibrated nozzles, controlled distance, and verified pressure/flow, then inspect and document any ingress points.

Hose-down test

Simulates cleaning/jet washing

Why it matters: Ensures wash-down jets won't force water past gaskets and cable glands—critical for industrial sites where cabinets are cleaned routinely. Example: preventing water-driven ingress into RCDs or PLC terminals.

How we run it: We run a water-jet rig: pump + calibrated nozzle, controlled distance/angle, verified pressure and flow (gauge + flowmeter), timed sweeps over seams/entries. After: open-up inspection and functional checks.

Indoor corrosion protection

Assesses indoor corrosion resistance

Why it matters: Confirms coatings resist indoor humidity/contaminants—preventing rust on door hardware and maintaining bonding/earth continuity over the product's service life.

How we run it: Exposure in controlled damp-heat/humidity chambers; temperature and RH are logged. Post-test: visual rating, check moving parts, and verify key interfaces and coatings.

Additional corrosion protection

Extra corrosion scenario check

Why it matters: Demonstrates extra margin for harsher industrial atmospheres—helping ensure long-term sealing and safe access even where chemicals or high humidity accelerate corrosion.

How we run it: Same corrosion rigs but with increased severity (longer duration/cycles or stronger conditions), with documented parameters and post-exposure ratings of corrosion, adhesion, and hardware operation.

Outdoor corrosion (salt spray 60h)

Simulates coastal/salty environments

Why it matters: Validates coastal/road-salt durability—critical because chlorides rapidly attack fasteners and edges, risking seized locks, degraded gaskets, and loss of IP protection.

How we run it: We expose in a salt spray chamber for the specified time (e.g., 60 h), controlling fog concentration, temperature, and pH; then assess corrosion sites, coating failure, and operability of doors/locks.

Oil exclusion

Checks protection against oil ingress

Why it matters: Ensures oily liquids won't creep through seals and attack insulation or gaskets—important near machinery where lubricants can soften elastomers and cause leaks.

How we run it: We apply oil to defined interfaces (door seal line, glands) for controlled time/temperature, using calibrated volumes; then inspect for ingress and check gasket swelling/softening (dimensional check + hardness if needed).

Oil immersion

Assesses material behavior in oil

Why it matters: Confirms materials and seals survive full oil contact without cracking or losing mechanical strength—preventing unexpected leaks or brittle failures in industrial environments.

How we run it: Parts/enclosure sections are immersed in an oil bath at controlled temperature; we measure before/after dimensions with calipers, check mass/appearance, and verify sealing and mechanical function.

Assembly & operation checks

Test title	Standard
Mechanical operation	IEC 62208
Lifting / handling	IEC 62208
Metal insert axial load	IEC 62208
Lock test	Internal Method
Cold bending	Internal Method

Test descriptions

Mechanical operation

Checks moving parts operate reliably

Why it matters: Ensures doors, latches, hinges, and mechanisms operate smoothly—so users can safely access equipment without forcing (which damages seals and hardware).

How we run it: Repeated operation on a cycle rig; forces measured with a force gauge, alignment checked with gauges, and closures verified for consistent engagement.

Lifting / handling

Checks safe lifting/handling behavior

Why it matters: Confirms lifting points/handles safely support the enclosure mass—preventing drops that could damage internal electrical assemblies.

How we run it: Load is applied using a hoist/test frame with controlled mass; we monitor deformation and attachment integrity and inspect after unloading for cracks or loosening.

Metal insert axial load

Checks insert strength under pull

Why it matters: Ensures embedded inserts won't pull out during assembly/maintenance, preventing loose mounting rails or unsafe device fixation.

How we run it: Axial pull is applied with a calibrated pull-out tester/universal testing machine; maximum force and failure mode are recorded.

Lock test

Checks lock function & reliability

Why it matters: Verifies lock reliability so the door stays securely closed—protecting IP and preventing unauthorized access.

How we run it: We cycle the lock with a defined number of operations; torque/operating force is checked with a torque meter/force gauge, and we inspect wear and alignment.

Cold bending

Checks flexibility at low temperature

Why it matters: Ensures plastics don’t become brittle in cold environments—so cable guides, covers, and accessories won’t crack during winter installation.

How we run it: Parts are conditioned in a cold chamber to the specified temperature, then bent over a defined mandrel/fixture; we inspect for cracks and loss of function.

Fire & heat behavior

Test title	Standard
Thermal stability	IEC 62208
Resistance to normal heat	IEC 62208
Resistance to abnormal heat & fire	IEC 62208
Flammability	UL

Test descriptions

Thermal stability

Checks shape stability under heat

Why it matters: Confirms the enclosure keeps shape under heat so doors still seal and internal mounting remains aligned—avoiding gaps that let dust/water in.

How we run it: We condition the enclosure/parts in a controlled thermal chamber/oven; temperatures are logged. Before/after, we check critical dimensions with calipers and verify door operation and gasket compression.

Resistance to normal heat

Checks durability at typical temperatures

Why it matters: Ensures materials tolerate typical operating temperatures near devices (breakers, power supplies) without softening or creeping—maintaining safety and IP.

How we run it: We expose the enclosure/parts in a temperature-controlled chamber for the specified duration, then check deformation, surface changes, and functional operation (doors/locks).

Resistance to abnormal heat & fire

Checks behavior in extreme conditions

Why it matters: Validates behavior in fault scenarios (overheating, arcing): materials should resist ignition and limit flame spread—helping contain an incident inside the cabinet.

How we run it: Depending on the requirement, we run standardized fire/heat methods (e.g., glow-wire or needle-flame) with controlled temperature and application time; we record afterflame/afterglow and inspect damage and dripping.

Flammability

Checks material reaction to flame

Why it matters: Ensures plastic parts self-extinguish and don't propagate fire—critical so the enclosure doesn't become fuel if an internal fault ignites nearby materials.

How we run it: We run a controlled flammability flame test on specimens using a calibrated burner, fixed flame time and angle; we measure burn length/time and dripping/afterflame.

Marking, appearance & usability

Test title	Standard
Marking check	IEC 62208
Conformity of marking	IEC 61386-1
Marking resistance	IEC 61386-1
Perceived quality check	Internal Method

Test descriptions

Marking check

Checks markings are present and readable

Why it matters: Guarantees installers can read ratings, wiring info, and warnings—preventing misapplication (wrong voltage/current) and unsafe installation.

How we run it: Visual verification under controlled lighting, using checklists; where needed, we measure label size/position with calipers/rulers and confirm permanence on representative samples.

Conformity of marking

Checks required markings are correct

Why it matters: Ensures markings match the standard and product configuration—so inspectors and customers can trust the declared ratings and approvals.

How we run it: Cross-check against standard requirements and technical documentation (ratings, symbols, IP/IK, material info); discrepancies are recorded as nonconformities.

Marking resistance

Checks labels resist handling/cleaning

Why it matters: Ensures labels/printing survive cleaning and handling—so safety info remains readable for maintenance throughout service life.

How we run it: We perform standardized rub/abrasion and cleaning-agent exposure using defined cloths/pressure/cycles; then evaluate legibility and adhesion (peeling, smearing).

Perceived quality check

Visual check: finish, defects, consistency

Why it matters: Confirms consistent finish and build quality—important because visible defects can hide functional issues (poor sealing surfaces, imperfect gasket seats).

How we run it: Structured visual inspection using acceptance criteria (surface, edges, coating uniformity), aided by controlled lighting and reference samples; findings are documented.

Electrical safety

Test title	Standard
Dielectric strength	IEC 62208
Earth continuity	IEC 62208

Test descriptions

Dielectric strength

Checks insulation withstands voltage

Why it matters: Confirms insulation barriers in the enclosure system withstand high voltage—preventing flashover and electric shock risk if a fault occurs. Example: ensuring a faulted conductor won't arc to the door or accessible metal parts.

How we run it: We use a calibrated hipot (dielectric withstand) tester to apply the specified voltage between conductive parts and accessible surfaces for a defined time; we monitor leakage/current and verify no breakdown.

Earth continuity

Checks grounding continuity

Why it matters: Ensures protective earth is reliable so fault current flows safely—reducing touch-voltage risk on metal parts (door, frame, gland plates).

How we run it: We measure low resistance using a calibrated micro-ohmmeter (often with high test current); probes are placed on designated points (door bond, frame, PE stud) and results recorded against limits.

Acoustics & noise

Test title	Standard
Sound power level measurement	ISO 13347-1

Test descriptions

Sound power level measurement

Quantifies noise output

Why it matters: Quantifies noise emissions independent of distance—useful to compare products and verify acoustic requirements for comfort/safety (e.g., cabinets with fans or installed devices that generate noise).

How we run it: We test in a controlled acoustic environment (e.g., semi-anechoic setup) using calibrated microphones and a sound level meter, with a calibrator before/after; we follow defined microphone positions and compute sound power per the method.

Laboratory Sites

Capellades (Barcelona, Spain)

Historic manufacturing site established in 1958 (HIMEL origin), today a key Schneider Electric hub for metal enclosures. Approx. 28,000 m² and 200+ industrial employees, with on-site laboratory capabilities supporting enclosure validation.

Molins de Rei (Barcelona, Spain)

Strategic production site for electrical cabinets and enclosures, with ongoing decarbonization initiatives (microgrid, renewable generation and storage). 160+ employees, producing PanelSet ranges for Spain and wider Europe, supported by lab and industrial innovation resources.

Sarel (Alsace, France)

Industrial site linked to the Sarel enclosure portfolio (brand founded 1956; integrated into Schneider Electric in 1984). Manufactures enclosures, cabinets and racks with integrated metalwork and finishing processes, complemented by continuous efficiency and emissions-reduction programs.